Senior Data Scientist & ML Mentor sifatida menga yordam ber

Men hozirgacha Data Analytics, Statistics, Probability, SQL, Python (Pandas, NumPy, Matplotlib, Seaborn, Plotly), Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Hypothesis Testing, Machine Learning fundamentals va Visualization bo'yicha o'rgangan barcha bilimlarimni namoyish qiladigan **bitta katta real-world portfolio loyihasi** yaratmoqchiman.

Bu loyiha mening:

- GitHub portfolioim,
- Kaggle portfolioim,
- Resume/CV,
- AI/Data Science internship va ish suhbatlari

uchun asosiy ("flagship") loyiha bo'ladi.

Loyiha mavzusi

Loyihaning asosiy maqsadi — **O'zbekistondagi eng katta e-commerce platformalaridan biri bo'lgan Uzum Market ma'lumotlarini chuqur Data Analytics va Business Intelligence nuqtai nazaridan tahlil qilish.**

Bu oddiy EDA loyihasi emas.

Men professional Data Analyst, Business Analyst va Junior Data Scientist darajasidagi ishlarni ko'rsatadigan loyiha yaratmoqchiman.

Biznes muammosi

Tasavvur qilaylik, men kelajakda Uzum Market platformasida o'zimning online do'konimni ochmoqchiman.

Bunga investitsiya qilishdan oldin quyidagi savollarga data orqali javob topishim kerak.

Masalan:

- O'zbekistonda online shopping qanchalik rivojlangan?
- Qaysi viloyatlarda qaysi mahsulotlarga talab yuqori?
- Talab mavsumlarga qarab qanday o'zgaradi?
- Bayramlarda yoki maxsus aksiyalarda qanday mahsulotlar ko'proq sotiladi?

- Narx va sotilish soni o'rtasida qanday bog'liqlik mavjud?
- Qaysi category eng tez o'smoqda?
- Qaysi mahsulotlarda raqobat yuqori?
- Qaysi mahsulotlarda bozorga kirish imkoniyati yaxshi?
- Xaridorlarning baholari va reviewlari sotuvga qanday ta'sir qiladi?
- Discount darajasi sotuvni qanchalik oshiradi?
- Qaysi product segmentlar risk jihatdan xavfsizroq?
- Inventory qanday rejalashtirilishi kerak?
- Qaysi davrlarda qaysi mahsulotni sotish eng foydali bo'ladi?
- Data asosida qanday biznes strategiya ishlab chiqish mumkin?

Maqsad — biznesdagi risklarni maksimal darajada kamaytirish va qarorlarni faqat ma'lumotlarga asoslangan holda qabul qilish.

Loyiha maqsadi

Bu loyiha orqali men quyidagilarni o'rganishni va namoyish qilishni xohlayman:

- Professional Data Analytics
- Business Analytics
- Statistical Analysis
- Exploratory Data Analysis
- Data Storytelling
- Machine Learning
- Forecasting
- Recommendation
- Dashboard Development
- Business Intelligence
- End-to-End Data Project Development

Men ishlatmoqchi bo'lgan texnologiyalar

Faqat Pandas bilan cheklanib qolishni istamayman.

Loyiha davomida imkon qadar ko'proq zamonaviy texnologiyalarni qo'llashni xohlayman.

Masalan:

- Python
- Pandas
- NumPy
- database: csv or sql
- BeautifulSoup

- Selenium yoki Playwright
- Matplotlib
- Seaborn
- Plotly
- Plotly Dash
- Streamlit
- Prophet
- Time Series Analysis
- ARIMA
- Clustering
- PCA
- Association Rules (Market Basket Analysis)
- Recommendation System
- NLP (Product Reviews)
- Sentiment Analysis
- Geospatial Analysis
- Feature Engineering
- Statistical Testing
- A/B Testing
- Docker
- Git
- GitHub Actions
- Kaggle Notebook
- Jupyter Notebook

Loyiha yakunida quyidagilar tayyor bo'lishi kerak

- Professional README
- Toza repository structure
- Modular Python code
- Data pipeline
- ETL pipeline
- Data Validation
- Exploratory Data Analysis
- Interactive Dashboard
- Machine Learning models
- Forecasting
- Business Insights
- Final Report
- Presentation
- Kaggle Notebook
- GitHub Portfolio
- Documentation

Sening vazifang

Sen 10+ yillik tajribaga ega Senior Data Scientist, Analytics Engineer va Business Intelligence Architect sifatida menga mentorlik qilasan.

Biz bu loyihani real kompaniya darajasidagi standartlar asosida ishlab chiqamiz.

Menga tayyor loyiha yozib berma.

Balki meni bosqichma-bosqich o'rgat.

Har bir qarorni nima uchun qabul qilayotganimizni tushuntir.

Har bir pipeline professional kompaniyalarda qanday ishlashini ko'rsat.

Har bir texnologiyani nima sababdan ishlatayotganimizni tushuntir.

Har bir bosqich tugagach uni review qil.

Code quality, architecture va best practicelarni ham o'rgat.

Birinchi vazifa

Avvalo ushbu loyiha uchun **to'liq End-to-End Project Roadmap** ishlab chiq.

Roadmap quyidagilarni o'z ichiga olishi kerak:

1. Project Architecture
2. Data Sources
3. Data Collection Strategy
4. Data Storage
5. ETL Pipeline
6. Data Validation
7. Data Cleaning
8. Feature Engineering
9. Exploratory Data Analysis
10. Statistical Analysis
11. Business Analytics
12. Customer Analytics
13. Product Analytics
14. Sales Analytics
15. Time Series Analysis
16. Forecasting

17. Machine Learning
18. Recommendation System
19. Dashboard
20. Deployment
21. Documentation
22. GitHub Structure
23. Kaggle Version
24. Final Presentation
25. Future Improvements

Har bir bo'lim uchun:

- Maqsad
- Nima uchun kerakligi
- Foydalaniladigan texnologiyalar
- Deliverables
- O'rganiladigan yangi mavzular
- Industry best practices
- Portfolio value
- GitHub commit strategiyasi

kabi bo'limlarni ham batafsil yoz.

Bu loyiha oddiy portfolio emas, balki Senior Data Analyst va Junior Data Scientist darajasidagi kuchli portfolio bo'lishi kerak.